Ratios, Proportions & Percentages

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Simplifying ratios**

Simplify the ratio $18 : 24$.

GCD of 18 and 24 is 6: $18 \div 6 : 24 \div 6 = 3 : 4$.

2 **[Foundational]** • **Target: Simplifying ratios**

Simplify the ratio $45 : 60 : 75$.

GCD of $45, 60, 75$ is 15: $3 : 4 : 5$.

3 **[Foundational]** • **Target: Simplifying ratios**

Express "12 boys to 18 girls" as a ratio in simplest form.

$12 : 18$, and GCD is 6, so $2 : 3$.

4 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

Share 150 in the ratio $2 : 3$.

$2 + 3 = 5$ parts, each part $= 150 \div 5 = 30$. Shares: 60 and 90.

5 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

Share 220 among three people in the ratio $3 : 4 : 4$.

$3 + 4 + 4 = 11$ parts, each part $= 220 \div 11 = 20$. Shares: 60, 80, 80.

6 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

A prize of 900 riyals is shared between Sara and Layla in the ratio $5 : 4$. Find each share.

$5 + 4 = 9$ parts, each part $= 900 \div 9 = 100$. Sara receives 500 riyals; Layla receives 400 riyals.

7 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

Divide 84 kg of flour in the ratio $1 : 2 : 4$.

$1 + 2 + 4 = 7$ parts, each part $= 84 \div 7 = 12$ kg. Shares: 12 kg, 24 kg, 48 kg.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Equivalent ratios and proportions

If $3:5 = x:20$, find x.

$\frac{3}{5} = \frac{x}{20}$, so $x = \frac{3 \times 20}{5} = 12$.

9 [Intermediate] • Target: Equivalent ratios and proportions

If $a:7 = 12:21$, find a.

$\frac{12}{21} = \frac{4}{7}$, so $a = 4$.

10 [Intermediate] • Target: Equivalent ratios and proportions

A map has scale $1:50{,}000$. A distance of 3 cm on the map represents how many meters in reality?

$3 \times 50{,}000 = 150{,}000$ cm $= 1{,}500$ m.

11 [Intermediate] • Target: Equivalent ratios and proportions

Solve for y: $\frac{4}{9} = \frac{y}{63}$.

$y = \frac{4 \times 63}{9} = 28$.

12 [Intermediate] • Target: Percent of a number

Find 35% of 240.

$\frac{35}{100} \times 240 = 84$.

13 [Intermediate] • Target: Percent of a number

What is 8% of 650?

$\frac{8}{100} \times 650 = 52$.

14 [Intermediate] • Target: Converting between fractions, decimals and percentages

Write $\frac{3}{8}$ as a percent.

$\frac{3}{8} = 0.375 = 37.5\%$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 3.01 Algebra Essentials

Write 0.45 as a fraction in lowest terms.

$0.45 = \frac{45}{100} = \frac{9}{20}$.

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16 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 4.01 Geometry & Measurement

Write 62.5% as a fraction in lowest terms.

$62.5\% = \frac{625}{1000} = \frac{5}{8}$.

17 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 5.01 Quantitative Comparisons

Write $\frac{7}{40}$ as a decimal.

$\frac{7}{40} = 0.175$.

18 [Challenging] • Target: Percentage increase and decrease • Cross-Lesson: 6.01 Sequences & Patterns

Increase 80 by 25%.

$80 \times 1.25 = 100$.

19 [Challenging] • Target: Percentage increase and decrease • Cross-Lesson: 7.01 Data Analysis & Probability

Decrease 150 by 20%.

$150 \times 0.8 = 120$.

20 [Challenging] • Target: Percentage increase and decrease • Cross-Lesson: 8.01 Word Problems & Reasoning

A salary of $4{,}000$ riyals increases by 6%. Find the new salary.

$4{,}000 \times 1.06 = 4{,}240$ riyals.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Sharing in a given ratio

The bar chart shows the ratio in which a total of 84 is to be shared. Find the larger share.

a. 36

b. 21

c. 48 ✓

d. 63

Total parts $= 3 + 4 = 7$; each part $= 84 \div 7 = 12$. Larger share $= 4 \times 12 = 48$.

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22 [Foundational] • Target: Percent of a number

The bar chart shows four students' quiz scores out of 80. Which score is exactly 75%?

a. 68

b. 64

c. 60 ✓

d. 56

75% of 80 = 0.75 × 80 = 60.

23 [Foundational] • Target: Percent of a number

The bar chart shows monthly rent for four apartments. 8% of which apartment's rent equals 200 SAR?

a. Apt. 2

b. Apt. 1

c. **Apt. 3** ✓

d. Apt. 4

0.08 × 2,500 = 200 SAR, matching Apartment 3's rent of 2,500 SAR.

24 [Foundational] • Target: Percentage increase & decrease

The bar chart shows a salary before and after a raise. What percentage increase does it represent?

a. 15%

b. 500%

c. 20% ✓

d. 25%

Increase = 6,000 − 5,000 = 1,000. Percentage increase = $\frac{1,000}{5,000}$ × 100% = 20%.

25 [Foundational] • Target: Successive percentage changes

The bar chart shows a stock's value at the start and after rising 50% in year one. It then falls 50% in year two. Find its final value.

a. 500

b. 900

c. 1,000

d. 750 ✓

1,000 × 1.5 × 0.5 = 750.

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26 [Foundational] • Target: Direct proportion

The graph shows the direct proportion between the number of hours x worked and pay y in SAR, $y = 15x$. How much is earned for 6 hours?

a. 105

b. 90 ✓

c. 75

d. 21

$y = 15 \times 6 = 90$ SAR.

27 [Foundational] • Target: Best value & unit price

The bar chart shows the price per 100 g for three brands of coffee. Which brand offers the best value?

a. **Brand B** ✓

b. All are equal

c. Brand C

d. Brand A

The lowest price per 100 g is the best value — Brand B at 9 SAR/100g beats Brand A (12) and Brand C (11).

28 [Foundational] • Target: Best value & unit price

The bar chart shows the total price of two cereal boxes of different sizes: 400 g and 600 g. Which has the lower price per 100 g?

a. Cannot be determined

b. They are equal

c. **The 600 g box** ✓

d. The 400 g box

400 g: $12 \div 4 = 3$ SAR/100g. 600 g: $16.50 \div 6 = 2.75$ SAR/100g. The 600 g box is cheaper per 100 g.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Word problems

The bar chart shows subject preference percentages from a survey of 250 students. How many students prefer math?

a. 60

b. 90

c. 110

d. 100 ✓

$0.40 \times 250 = 100$ students.

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30 [Intermediate] • Target: Simplifying & writing ratios

Simplify $18 : 45$.

a. $3 : 5$

b. $2 : 9$

c. $2 : 5$ ✓

d. $6 : 15$

GCD of 18 and 45 is 9: $18 \div 9 : 45 \div 9 = 2 : 5$.

31 [Intermediate] • Target: Simplifying & writing ratios

A recipe uses 250 g flour and 2 eggs. Write the ratio of flour (in grams) to eggs in simplest form.

a. $1 : 125$

b. $250 : 2$

c. $125 : 1$ ✓

d. $2 : 250$

$250 : 2$ simplifies by dividing both by 2: $125 : 1$.

32 [Intermediate] • Target: Simplifying & writing ratios

Express $0.75 : 2$ as a ratio of whole numbers in simplest form.

a. $3 : 4$

b. $3 : 8$ ✓

c. $75 : 2$

d. $75 : 200$

Multiply both terms by 4 to clear the decimal: $3 : 8$.

33 [Intermediate] • Target: Sharing in a given ratio

Share 260 SAR in the ratio $2 : 3 : 5$. Find the smallest share.

a. 78 SAR

b. 52 **SAR** ✓

c. 130 SAR

d. 65 SAR

Total parts $= 2 + 3 + 5 = 10$; each part $= 260 \div 10 = 26$. Smallest share $= 2 \times 26 = 52$.

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34 [Intermediate] • Target: Sharing in a given ratio

Two business partners share a profit of $18{,}000$ SAR in the ratio $5:7$. Find the larger share.

a. $10{,}500$ **SAR** ✓

b. $9{,}000$ SAR

c. $7{,}500$ SAR

d. $12{,}857$ SAR

Each part $= 18{,}000 \div 12 = 1{,}500$. Larger share $= 7 \times 1{,}500 = 10{,}500$ SAR.

35 [Intermediate] • Target: Sharing in a given ratio

A line of length 36 cm is divided in the ratio $4:5$. Find the length of the shorter piece.

a. 16 ✓

b. 4

c. 9

d. 20

Total parts $= 4 + 5 = 9$; each part $= 36 \div 9 = 4$ cm. Shorter piece $= 4 \times 4 = 16$ cm.

36 [Intermediate] • Target: Sharing in a given ratio

A class of 40 students has boys and girls in the ratio $3:5$. How many boys are there?

a. 25

b. 8

c. 24

d. 15 ✓

Total parts $= 3 + 5 = 8$; each part $= 40 \div 8 = 5$. Boys $= 3 \times 5 = 15$.

37 [Intermediate] • Target: Equivalent ratios & scale

If $a : b = 3 : 5$ and $a = 18$, find b.

a. 30 ✓

b. 15

c. 10.8

d. 24

Scale factor $= 18 \div 3 = 6$. $b = 5 \times 6 = 30$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Equivalent ratios & scale • Cross-Lesson: 6.01 Sequences & Patterns

A map has scale $1:50{,}000$. A distance of 3 cm on the map represents how many km in reality?

a. 15 km

b. 1.5 **km** ✓

c. 150 km

d. 0.15 km

$3 \times 50{,}000 = 150{,}000$ cm $= 1{,}500$ m $= 1.5$ km.

39 [Challenging] • Target: Equivalent ratios & scale • Cross-Lesson: 7.01 Data Analysis & Probability

Which ratio is equivalent to $4:9$?

a. $9:4$

b. $20:45$ ✓

c. $8:16$

d. $16:37$

$4:9$ scaled by 5 gives $20:45$.

40 [Challenging] • Target: Equivalent ratios & scale • Cross-Lesson: 8.01 Word Problems & Reasoning

In a photo, a model that is 1.8 m tall appears 6 cm high. Using the same scale, how tall (in m) is a building that appears 40 cm high in the photo?

a. 4.8

b. 24

c. 7.2

d. 12 ✓

Scale: 6 cm represents 1.8 m, so 1 cm represents 0.3 m. $40 \times 0.3 = 12$ m.

41 [Challenging] • Target: Percent of a number • Cross-Lesson: 1.01 Arithmetic & Number Sense

150% of a number is 300. Find the number.

a. 250

b. 450

c. 200 ✓

d. 150

$1.5 \times x = 300 \Rightarrow x = 300 \div 1.5 = 200$.

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42 [Challenging] • Target: Fraction, decimal & percent conversion • Cross-Lesson: 3.01 Algebra Essentials

Write 0.06 as a percent.

a. 60%

b. 6% ✓

c. 0.06%

d. 0.6%

Move the decimal point two places right: $0.06 = 6\%$.

43 [Challenging] • Target: Fraction, decimal & percent conversion • Cross-Lesson: 4.01 Geometry & Measurement

Write 85% as a fraction in simplest form.

a. $\frac{17}{20}$ ✓

b. $\frac{8}{5}$

c. $\frac{17}{25}$

d. $\frac{85}{100}$

$85\% = \frac{85}{100}$, which simplifies (divide by 5) to $\frac{17}{20}$.

44 [Challenging] • Target: Fraction, decimal & percent conversion • Cross-Lesson: 5.01 Quantitative Comparisons

Which of these is largest: $\frac{3}{8}$, 0.4, or 37%?

a. 0.4 ✓

b. They are all equal

c. $\frac{3}{8}$

d. 37%

$\frac{3}{8} = 0.375 = 37.5\%$, and $37\% = 0.37$. So 0.4 is the largest.

45 [Challenging] • Target: Percentage increase & decrease • Cross-Lesson: 6.01 Sequences & Patterns

A shirt costs 120 SAR after a 20% discount. Find the original price.

a. 150 **SAR** ✓

b. 144 SAR

c. 100 SAR

d. 140 SAR

$120 = 0.8 \times \text{original} \Rightarrow \text{original} = 120 \div 0.8 = 150$ SAR.

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